

RED DOG MINE 3 SSW, AK, USA

WMO: 701220

Lat: 68.0277N

Lon: 162.9212W

Elev: 287

StdP: 97.92

Time Zone: -9.00 (NAK)

Period: 10-19

WBAN: 26655

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.5	-29.5	-35.9	0.1	-32.1	-33.2	0.2	-29.0	11.3	-13.0	10.3	-11.4	1.3	N/A	0.776

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	22.8	13.6	20.5	12.7	18.6	12.0	14.6	20.3	13.7	18.7	12.8	17.3	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.5	9.3	15.4	11.4	8.7	14.9	10.4	8.1	14.0	41.6	19.8	39.1	18.9	36.8	17.3	17.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.8	6.7	-32.5	26.1	4.1	2.4	-35.4	27.9	-37.8	29.2	-40.1	30.6	-43.0	32.3		
(5)			-32.5	16.5	3.7	0.8	-35.2	17.1	-37.4	17.6	-39.4	18.1	-42.1	18.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-14.1	-13.2	-14.0	-6.5	2.5	10.5	11.7	9.0	4.1	-2.9	-10.6	-14.6
	DBStd	11.60	9.40	7.94	6.39	5.81	6.14	4.46	3.58	3.50	4.55	4.60	5.94	7.04
	HDD10.0	4966	747	649	743	496	239	48	21	59	182	399	618	763
	HDD18.3	7833	1006	883	1001	746	489	237	207	288	428	658	868	1022
	CDD10.0	179	0	0	0	0	8	63	75	29	4	0	0	0
	CDD18.3	4	0	0	0	0	0	2	2	0	0	0	0	0
	CDH23.3	48	0	0	0	0	2	24	23	0	0	0	0	0
	CDH26.7	7	0	0	0	0	0	5	2	0	0	0	0	0
(14) Wind	WSAvg	2.6	3.0	3.4	2.7	2.9	2.5	2.2	2.0	2.1	2.3	2.5	2.9	3.4
(15) Precipitation	PrecAvg	452	32	35	26	21	17	20	49	72	50	60	39	31
	PrecMax	630	68	111	126	55	46	37	98	155	111	116	116	78
	PrecMin	324	7	5	6	3	1	9	12	14	13	24	6	10
	PrecStd	79	16	26	24	15	12	8	20	33	21	24	25	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	2.0	1.0	7.8	21.0	26.5	26.5	21.8	16.5	11.2	3.2	1.9
		MCWB	2.0	-0.2	-1.5	3.6	11.3	15.4	15.1	13.0	9.6	3.9	1.2	-0.1
	2%	DB	0.7	-0.2	-0.6	4.5	17.0	23.0	22.9	18.9	14.2	7.5	1.2	0.3
		MCWB	-0.9	-1.7	-1.9	1.2	9.7	13.0	14.5	12.3	8.5	2.5	-0.2	-1.2
	5%	DB	-0.8	-1.5	-2.5	2.7	13.9	20.6	20.6	16.5	12.3	5.0	-0.5	-1.7
		MCWB	-2.0	-2.5	-3.5	0.3	8.1	12.1	13.5	11.4	8.0	2.3	-1.5	-3.1
	10%	DB	-2.4	-2.9	-4.3	1.1	11.0	18.7	18.6	14.8	10.5	3.2	-2.3	-4.3
		MCWB	-3.7	-3.8	-5.1	-0.7	6.3	11.2	12.9	10.6	7.6	1.8	-3.2	-5.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	0.3	-0.4	3.6	11.8	15.9	16.2	14.1	10.8	4.8	1.7	0.7
		MCDB	4.5	1.2	0.1	7.7	20.5	24.8	23.7	17.0	13.4	7.5	3.0	1.3
	2%	WB	-0.3	-1.4	-2.0	1.4	9.8	14.0	15.2	13.3	9.9	3.7	0.1	-1.1
		MCDB	0.5	-0.6	-1.1	3.9	16.9	20.8	21.3	17.1	12.3	5.9	1.1	0.4
	5%	WB	-1.9	-2.4	-3.5	0.5	8.4	13.0	14.2	12.4	8.9	2.7	-1.3	-3.1
		MCDB	-0.9	-1.6	-2.4	2.4	13.3	18.4	19.3	16.0	11.2	4.6	-0.4	-1.8
	10%	WB	-3.6	-3.7	-5.1	-0.5	6.8	12.0	13.4	11.3	8.0	1.8	-3.2	-5.0
		MCDB	-2.5	-3.0	-4.1	1.2	10.5	17.4	17.9	14.0	10.2	3.5	-2.4	-4.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.4	6.4	6.4	7.7	10.1	8.3	7.0	6.5	4.7	4.8	5.6
		MCDBR	4.8	5.5	6.2	6.7	12.1	13.3	12.5	9.7	8.5	5.5	4.0	5.3
	5% WB	MCWBR	3.9	5.1	5.5	4.7	6.5	6.1	5.8	5.0	4.6	3.3	3.5	4.6
		MCDWR	4.9	5.9	5.9	6.4	11.3	11.1	10.7	8.0	6.6	4.5	4.0	5.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.169	0.233	0.256	0.319	0.356	0.336	0.338	0.341	0.304	0.231	0.162	N/A
		taud	1.901	2.074	2.222	2.092	2.158	2.438	2.455	2.448	2.476	2.266	1.922	N/A
	Edn at Noon	Edn at Noon	126	571	797	818	826	867	852	804	748	592	117	0
		Edh at Noon	8	47	74	115	122	95	91	82	61	37	7	0
(44) All-Sky Solar Radiation	RadAvg	0.04	0.54	2.27	4.50	5.48	5.88	4.72	3.40	2.05	0.72	0.11	0.00	
	RadStd	0.01	0.06	0.17	0.23	0.60	0.51	0.53	0.42	0.23	0.08	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page